

Four routes to learned genomic representations

Source

Model input

Learning

Representation

a Linear sequence models

DNABERT · NT



Sequence

Tokens

Sequence
self-supervision



Sequence
embedding

b Graph-derived sequence models

DeepGene



Pangenome graph

Linearized
sequence

Sequence
learning



Sequence
embedding

c Task-specific graph models

PangenomeX



Population
relationships

CNV
network

Task
supervision



Graph
embedding

d Topology-native pretraining

PangenomeFM



Pangenome graph

Oriented
native graph

Masked
adjacency



Graph
embedding